
AI-Assisted Penetration Testing

Continuous security assessment of APIs, endpoints, domains, and network zones

Using the **AccuKnox AgentZ** Zero Trust agent platform

Attack surfaces are no longer static. APIs ship weekly, endpoints spin up and down, domains multiply, and cloud network zones shift by the hour. A point-in-time pen test, run once a quarter by scarce human specialists, cannot keep pace. **AccuKnox AI-Assisted Penetration Testing** replaces the periodic engagement with continuous, autonomous adversarial assessment across your APIs, applications, endpoints, domains, and network zones, with every testing action governed inside a sandbox you control.

24/7

Continuous, autonomous
assessment

0

Standing agent privileges

100%

Actions logged and
policy-governed

1

Unified platform for the
whole estate

THE CHALLENGE

Why traditional pen testing breaks down

Security teams defend an estate that changes faster than they can test it by hand. Three gaps compound.

Coverage gaps

A manual test samples a slice of the estate at a single point in time. API endpoints, subdomains, and network zones provisioned the next day go untested until the next engagement, often months later.

Speed and cost

Skilled testers are scarce and expensive. Backlogs mean critical exposures sit undiscovered while attackers scan continuously.

APIs move fastest of all

Every release adds routes, parameters, and versions. Undocumented and shadow APIs never reach a test scope, so broken authorization and overexposed data stay live until someone finds them.

The window between an exposure appearing and anyone finding it keeps widening. Closing it takes assessment that is continuous, automated, and safe to run against production-adjacent assets.

THE SOLUTION

Always-on assessment, using AgentZ

AccuKnox runs penetration testing as an always-on capability. Autonomous testing agents run using **AgentZ**, the AccuKnox Zero Trust sandbox platform. AgentZ gives each agent the skills, tool access, and workflows it needs to probe a target, while enforcing isolation, ephemeral credentials, and policy mediation so the assessment never becomes the incident.

Asset domain	Continuous assessment coverage
APIs	Discovery of documented, undocumented, and shadow endpoints, then authenticated and unauthenticated probing for broken object and function level authorization, injection, and data overexposure.
Web applications	Crawl and active testing of public and logged-in surfaces, baseline or full depth, against injection, cross-site scripting, header misconfiguration, and session flaws.
Domains	Subdomain and attack-surface discovery, plus exposed-service enumeration across the external footprint.
Endpoints	Workstations, servers, and containers tested for exploitable vulnerabilities, misconfiguration, weak credentials, and exposed services as they come online.
Network zones	Segmentation and lateral-movement testing across cloud and on-prem zones, confirming trust boundaries hold under real attack paths.
Cloud and Kubernetes	Entitlement, workload, and configuration exposure mapped and exercised against real exploit paths.

HOW IT WORKS

The assessment loop

A closed, repeating loop. Every stage runs inside the AgentZ sandbox, so autonomy never costs you control.

- 01 Discover**
 Agents map the live attack surface, so nothing is tested against a stale inventory.
- 02 Prioritize**
 Findings are scored by exploitability and business exposure, aligned with **CTEM**.
- 03 Simulate and exploit**
 Agents reproduce real adversary technique under controlled, isolated conditions.
- 04 Validate and report**
 Every confirmed exposure ships with evidence, location, and reproduction steps.
- 05 Re-test**
 After a fix, agents re-run the tests on a schedule to confirm closure.

Parameters

Target URL * ⓘ

https://ctflearn.com/

Scan Type * ⓘ

Baseline Scan

Quick Passive Scan

5-10 Minutes

Full Scan

Comprehensive Active Scan

30-60 Minutes

Authentication * ⓘ

No Authentication

Scan Publicly Accessible Pages Without Authentication

Authentication

Scan Secured Pages That Require User Login Credentials

Scope a target: URL, baseline or full depth, and whether the agent tests authenticated surfaces.

☐	Last Seen	Vulnerability Name	Asset Name	Description	Risk Factor	Location	A
☐	2025-11-14 11:04:00	Re-examine Cache-control Dir	https://accuknox.com	The cache-control header has i	Informational	https://accuknox.com/	V
☐	2025-11-14 11:04:00	Insufficient Site Isolation Again	https://accuknox.com	Cross-Origin-Embedder-Policy	Low	https://accuknox.com/	V
☐	2025-11-14 11:04:00	Sub Resource Integrity Attribut	https://accuknox.com	The integrity attribute is missin	Medium	https://accuknox.com/	V
☐	2025-12-05 15:38:54	Content Security Policy (CSP)	https://www.accuknox.com	Content Security Policy (CSP)	Medium	GET, https://www.accuknox.cc	V
☐	2025-11-14 11:04:00	Content Security Policy (CSP)	https://accuknox.com	Content Security Policy (CSP)	Medium	https://accuknox.com/	V
☐	2025-11-14 11:04:00	Timestamp Disclosure - Unix	https://accuknox.com	A timestamp was disclosed by	Low	https://accuknox.com/	V
☐	2025-11-14 11:04:00	Information Disclosure - Suspici	https://accuknox.com	The response appears to conte	Informational	https://accuknox.com/	V

Findings carry risk factor, affected location, asset type, and the fix.

BUILT ON ZERO TRUST

Why autonomous testing stays safe to run

Handing an agent the ability to run exploits is only responsible if that agent is contained. AgentZ contains it four ways.

Sandbox isolation

Process, filesystem, credential, and network isolation enforced at the kernel by KubeArmor, eBPF, and Linux Security Modules.

Zero standing privilege

Ephemeral, just-in-time credentials scoped to one task, so a compromised agent has nothing durable to abuse.

Policy mediation

Real-time validation inspects every action and tool call before it leaves the sandbox.

Full auditability

Every agent action is logged and governed, producing the evidence trail a compliance review asks for.

BUSINESS OUTCOMES

What security teams gain

Close the exposure window

Weaknesses surface as assets appear, not months later at the next engagement.

Test every API, not a sample

Discovery keeps the scope current, so new endpoints get probed in the same pass.

Scale scarce expertise

Agents take breadth and repetition, freeing experts for creative testing.

Prioritize what matters

CTEM-aligned scoring focuses remediation on what an attacker could reach.

Prove control, not just effort

Automatic re-testing verifies fixes and makes pen testing a validated control.

Stay audit-ready

Logged agent activity gives auditors a dated record of what was tested and fixed.

WHERE IT FITS

One capability in the AccuKnox platform

API Security

ASPM

CTEM

CSPM

CWPP

KSPM

Compliance

See continuous pen testing against your own estate

Book a demo and we will map your APIs, applications, endpoints, domains, and network zones to a live assessment loop, then walk the findings and the re-test schedule with your team.

[Book a demo](#)
[No-auth scan setup](#)
[Authenticated scan setup](#)
[API security guide](#)